

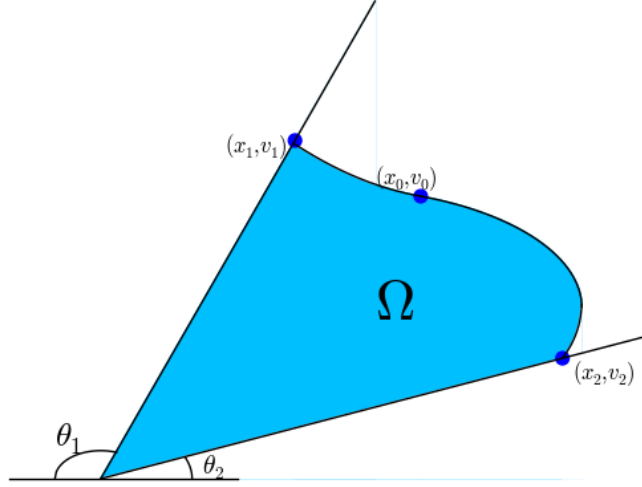


Figure 9:

which implies that:

$$x_m = \frac{r_1 \tan \theta_1 - r_2 \tan \theta_2 - (u_1 - u_2)}{\tan \theta_1 + \tan \theta_2} \quad (130)$$

Then we use x_m and ψ_m to express the total volume. By using geometric computation, we have:

$$\begin{aligned} \mathcal{V}(\psi_m) = & (r_1 u_1 - \int_{\psi_1}^{\psi_m} \int_{\psi}^{\psi_m} \frac{\cos \gamma}{g \sqrt{\frac{2\sigma}{g} (\cos \psi_m - \cos \gamma)}} d\gamma \frac{-\sin \psi}{g \sqrt{\frac{2\sigma}{g} (\cos \psi_m - \cos \psi)}} d\psi) \\ & + \int_{\psi_2}^{\psi_m} \int_{\psi}^{\psi_m} \frac{\cos \gamma}{g \sqrt{\frac{2\sigma}{g} (\cos \psi_m - \cos \gamma)}} d\gamma \frac{-\sin \psi}{g \sqrt{\frac{2\sigma}{g} (\cos \psi_m - \cos \psi)}} d\psi \\ & + \frac{1}{2} (u_1 + u_2)^2 \frac{1}{-\tan \theta_1} + (-u_2) r_1 + \frac{1}{2} (u_2 - u_1 - x_1 \tan \theta_1)^2 \left(\frac{1}{\tan \theta_1} + \frac{1}{\tan \theta_2} \right) \end{aligned} \quad (131)$$

Then we need to prove the following theorem about the total volume:

Theorem 3.12. *When $\psi_m \rightarrow 0$, we have $\mathcal{V}(\psi_m) \rightarrow +\infty$.*

Proof. Using the computation in Theorem 3.7, we know that:

$$\int_{\psi}^{\psi_m} \frac{\cos \gamma}{g \sqrt{\frac{2\sigma}{g} (\cos \psi_m - \cos \gamma)}} d\gamma \rightarrow +\infty$$